

Before class: energy and speed

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 15 minutes. Read the four short cards, then try the readiness check. Wrong answers are fine: they show what to listen for in the lecture.

Read the cards first, then answer the questions without notes. A periodic table and the answer key are at the back.

1 · Read first (5 min)

Bonds and energy (recap)

From Energy & Matters: **breaking** a bond **absorbs** energy and **forming** a bond **releases** energy. If more energy is released than absorbed, the reaction gives out heat: it is **exothermic** and ΔH is negative. If more is absorbed, it is **endothermic** and ΔH is positive.

Can it happen, and how fast?

Wood, paper and glucose all burn exothermically, yet none of them catches fire on your desk. Reactants must first climb an energy hill called the **activation energy** (E_a). The bigger the hill, the slower the reaction.

Particles must meet

Particles react only when they **collide**, with **enough energy** and the **right way round**. Anything that makes useful collisions more frequent (more particles, higher temperature, smaller pieces, a catalyst) makes the reaction faster. This is why medicines are kept cool, dry and tightly closed.

Reactions that go both ways

Some reactions are **reversible**: the products can turn back into reactants. We write a double arrow, $\rightleftharpoons$. Haemoglobin picking up oxygen in the lungs and releasing it in the tissues is one example you will meet today.

2 · Readiness check

Ten quick questions on what you need for today.

Q1 CR2-LO1

Breaking a chemical bond releases energy.

True / False

Q2 CR2-LO1

A reaction has $\Delta H = -890$ kJ. It is:

- A. endothermic: heat is absorbed
- B. exothermic: heat is released
- C. neither: no heat change
- D. impossible to tell

Q3 CR2-LO1

Match each process to its type.

Options: exothermic · endothermic

Burning natural gas

Ice melting

Photosynthesis

A hand warmer heating up

Q4 CR2-LO1

Recap: the molar mass of methane, CH_4 (C = 12.01, H = 1.01).

Q5 CR2-LO1

Recap: how many moles are in 32.1 g of CH₄ (16.05 g/mol)?

Q6 CR2-LO2

Paper releases a lot of energy when it burns, but it does not burst into flame on its own. The best reason is:

- A. it is endothermic
- B. it needs energy to start the reaction (activation energy)
- C. it contains no oxygen
- D. the bonds in paper are very weak

Q7 CR2-LO3

Every collision between reactant particles produces products.

True / False

Q8 CR2-LO4

Insulin is stored in the fridge mainly because:

- A. cold makes it more concentrated
- B. low temperature slows down the reactions that break it down
- C. cold kills bacteria in the vial
- D. insulin is a gas at room temperature

Q9 CR2-LO5

A substance that speeds up a reaction without being used up is called a

_____.

Q10 CR2-LO6

The double arrow $\rightleftharpoons$ in an equation means the reaction:

- A. is very fast
- B. is exothermic
- C. can go in both directions
- D. needs a catalyst

3 • Think ahead

No wrong answers here. Make a guess, then check it after the lecture (Parts 4 and 5).

Q11 CR2-LO4

A bottle of aspirin smells of vinegar after months in a hot, steamy bathroom. What do you think has happened, and why did the bathroom make it worse?

Q12 CR2-LO5

Hydrogen peroxide sits quietly in its bottle, but fizzes at once when you put it on a cut. Suggest why.

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Adapted from *The Basics of GOB Chemistry* (Ball, Hill & Scott), LibreTexts, CC BY-NC-SA 3.0, sections 6.5 and 7.4, with rates and equilibrium expanded from *Introductory Chemistry* (LibreTexts), chapter 15. Shared under the same licence for non-commercial teaching.

Periodic table · each box: atomic number, symbol, atomic mass, **electronegativity** · colours: metal, metalloid, nonmetal, noble gas

1A 1	2A 2	3	4	5	6	7	8	9	10	11	12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.01 2.1																	2 He 4.00 —
3 Li 6.94 1.0	4 Be 9.01 1.5											5 B 10.81 2.0	6 C 12.01 2.5	7 N 14.01 3.0	8 O 16.00 3.5	9 F 19.00 4.0	10 Ne 20.18 —
11 Na 22.99 0.9	12 Mg 24.31 1.2											13 Al 26.98 1.5	14 Si 28.09 1.8	15 P 30.97 2.1	16 S 32.07 2.5	17 Cl 35.45 3.0	18 Ar 39.95 —
19 K 39.10 0.8	20 Ca 40.08 1.0	21 Sc 44.96 1.3	22 Ti 47.87 1.5	23 V 50.94 1.6	24 Cr 52.00 1.6	25 Mn 54.94 1.5	26 Fe 55.84 1.8	27 Co 58.93 1.9	28 Ni 58.69 1.9	29 Cu 63.55 1.9	30 Zn 65.38 1.6	31 Ga 69.72 1.6	32 Ge 72.63 1.8	33 As 74.92 2.0	34 Se 78.97 2.4	35 Br 79.90 2.8	36 Kr 83.80 —
37 Rb 85.47 0.8	38 Sr 87.62 1.0	39 Y 88.91 1.2	40 Zr 91.22 1.4	41 Nb 92.91 1.6	42 Mo 95.95 1.8	43 Tc [98] 1.9	44 Ru 101.07 2.2	45 Rh 102.91 2.2	46 Pd 106.42 2.2	47 Ag 107.87 1.9	48 Cd 112.41 1.7	49 In 114.82 1.7	50 Sn 118.71 1.8	51 Sb 121.76 1.9	52 Te 127.60 2.1	53 I 126.90 2.5	54 Xe 131.29 —
55 Cs 132.91 0.7	56 Ba 137.33 0.9	57–71	72 Hf 178.49 1.3	73 Ta 180.95 1.5	74 W 183.84 1.7	75 Re 186.21 1.9	76 Os 190.23 2.2	77 Ir 192.22 2.2	78 Pt 195.08 2.2	79 Au 196.97 2.4	80 Hg 200.59 1.9	81 Tl 204.38 1.8	82 Pb 207.2 1.9	83 Bi 208.98 1.9	84 Po [209] 2.0	85 At [210] 2.2	86 Rn [222] —
87 Fr [223] 0.7	88 Ra [226] 0.9	89–103	104 Rf [267] —	105 Db [268] —	106 Sg [269] —	107 Bh [270] —	108 Hs [277] —	109 Mt [278] —	110 Ds [281] —	111 Rg [282] —	112 Cn [285] —	113 Nh [286] —	114 Fl [289] —	115 Mc [290] —	116 Lv [293] —	117 Ts [294] —	118 Og [294] —
Lanthanides		57 La 138.91 —	58 Ce 140.12 —	59 Pr 140.91 —	60 Nd 144.24 —	61 Pm [145] —	62 Sm 150.36 —	63 Eu 151.96 —	64 Gd 157.25 —	65 Tb 158.93 —	66 Dy 162.50 —	67 Ho 164.93 —	68 Er 167.26 —	69 Tm 168.93 —	70 Yb 173.05 —	71 Lu 174.97 —	
Actinides		89 Ac [227] 1.1	90 Th 232.04 1.3	91 Pa 231.04 1.4	92 U 238.03 1.4	93 Np [237] —	94 Pu [244] —	95 Am [243] —	96 Cm [247] —	97 Bk [247] —	98 Cf [251] —	99 Es [252] —	100 Fm [257] —	101 Md [258] —	102 No [259] —	103 Lr [262] —	

Atomic masses to 2 d.p. ([n] = mass number of the most stable isotope). Electronegativity: Pauling values as in Ball, *The Basics of GOB Chemistry*, Figure 4.4.3 (La–Lu 1.0–1.2; noble gases have no value).

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About 15 minutes. Read the four short cards, then try the readiness check. Wrong answers are fine: they show what to listen for in the lecture.

Answer key

2 · Readiness check

Q1 CR2-LO1

Breaking a chemical bond releases energy.

True / False → **False**

Breaking a bond **absorbs** energy; forming a bond releases it.

Q2 CR2-LO1

A reaction has $\Delta H = -890$ kJ. It is:

A. endothermic: heat is absorbed

B. exothermic: heat is released

C. neither: no heat change

D. impossible to tell

A negative ΔH means heat leaves the reaction: exothermic.

Q3 CR2-LO1

Match each process to its type.

Burning natural gas → exothermic; Ice melting → endothermic;

Photosynthesis → endothermic; A hand warmer heating up → exothermic

Heat given out = exothermic; heat taken in = endothermic.

Q4 CR2-LO1

Recap: the molar mass of methane, CH₄ (C = 12.01, H = 1.01).

16.05 g/mol

$12.01 + 4 \times 1.01 = 16.05$ g/mol. You will turn grams into moles, then moles into kJ.

Q5 CR2-LO1

Recap: how many moles are in 32.1 g of CH₄ (16.05 g/mol)?

2 mol

$32.1 \div 16.05 = 2.00$ mol.

Q6 CR2-LO2

Paper releases a lot of energy when it burns, but it does not burst into flame on its own. The best reason is:

- A. it is endothermic
- B. it needs energy to start the reaction (activation energy)**
- C. it contains no oxygen
- D. the bonds in paper are very weak

Even exothermic reactions must first get over an energy barrier.

Q7 CR2-LO3

Every collision between reactant particles produces products.

True / False → **False**

Only collisions with enough energy and the right orientation are effective.

Q8 CR2-LO4

Insulin is stored in the fridge mainly because:

- A. cold makes it more concentrated
- B. low temperature slows down the reactions that break it down**
- C. cold kills bacteria in the vial
- D. insulin is a gas at room temperature

Lower temperature means slower degradation reactions.

Q9 CR2-LO5

A substance that speeds up a reaction without being used up is called a _____.

catalyst

Enzymes are the catalysts of the body.

Q10 CR2-LO6

The double arrow $\rightleftharpoons$ in an equation means the reaction:

- A. is very fast
- B. is exothermic
- C. can go in both directions**
- D. needs a catalyst

$\rightleftharpoons$ shows a reversible reaction: forward and reverse.

3 · Think ahead

Q11 CR2-LO4

A bottle of aspirin smells of vinegar after months in a hot, steamy bathroom. What do you think has happened, and why did the bathroom make it worse?

Aspirin has reacted with water (hydrolysis) to form salicylic acid and acetic acid, which smells of vinegar. Heat speeds up the reaction (more collisions, more with energy $\geq E_a$) and steam supplies more water. You will see this on the slide
Pharmacy focus: why medicines expire.

Q12 CR2-LO5

Hydrogen peroxide sits quietly in its bottle, but fizzes at once when you put it on a cut. Suggest why.

Blood and tissue contain the enzyme **catalase**, a catalyst that lowers E_a for $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$. The fizz is oxygen gas. See Part 5 of the slides.

Adapted from *The Basics of GOB Chemistry* (Ball, Hill & Scott), LibreTexts, CC BY-NC-SA 3.0, sections 6.S and 7.4, with rates and equilibrium expanded from *Introductory Chemistry* (LibreTexts), chapter 15. Shared under the same licence for non-commercial teaching.